Accurate Closed-Form Approximation for Symbol Error Probability in Hexagonal QAM

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***Abstract*—Future communication systems are anticipated to facilitate applications requiring high data transmission rates while maintaining energy efficiency. Hexagonal quadrature amplitude modulation (HQAM) offers this owing to its compact symbol arrangement within the two-dimensional (2D) plane. Due to the structure of its hexagonal lattice, accurately calculating the error probability is highly challenging, if not entirely impractical. Building on the limitations of the current approaches, this letter presents a straightforward and precise approximation for calculating the symbol error probability (SEP) of HQAM in an additive white Gaussian noise channel. Simulations show the proposed approximation is highly accurate for all hexagonal QAM constellations. A novel cluster-based detection method with low computational complexity is proposed. The method ensures that its computational demand remains unaffected by increase in constellation order. Additionally, the average SEP performance of the system is evaluated under a practical scenario with imperfect channel state information over Nakagami-*m* fading channels.**

***Index Terms*—AWGN, HQAM, SEP, detection method, Nakagami-m, imperfect CSI.**

I. Introduction

With the global deployment of 5G and research on 6G focusing on the high-data-rate multimedia applications with minimal energy consumption. There is a need for optimizing bandwidth and power efficiency. Energy-efficient high-speed communication networks in future wireless systems involve minimizing the average transmit power while ensuring a certain bit error rate (BER) or symbol error rate (SER). In order to maintain SER performance at higher data rates within the available bandwidth, it typically becomes necessary to boost transmit power. The future need has led to research on higher-order two-dimensional (2D) constellations, which offer high data rates in an energy-efficient manner, as well as more power-efficient 2D hexagonal-shaped constellations, known as hexagonal quadrature amplitude modulation (HQAM) [1]. In a 2D signal constellation, the SER is primarily influenced by the minimum distance between neighboring constellation points and the average symbol energy, which relates to the mean squared distance of the points from the origin. Taking these factors into account, an optimal 2D hexagonal lattice-based HQAM constellation is suggested. In HQAM, if the minimum distance between, two adjacent points is d, the area of the hexagonal region is $\frac{\sqrt{3}}{2}d^2$. This area is 0.866 times that of a rectangular region with the same dimensions, resulting in an approximate gain of 0.6 dB over rectangular regions [1]. HQAM provides the most compact 2D packing, which lowers both the peak and average power of the constellation, thereby improving its power efficiency compared to other constellations. A key metric for assessing the performance of a modulation scheme is symbol error probability (SEP). However, the hexagonal lattice structure of HQAM makes it challenging to compute the exact SEP. Most works consider existing SEP approximations of HQAM to analyze system performance, as seen in [2], [3], [4], [5] and the references therein. However, only a few studies focus on the accuracy and computational complexity of these approximations. The SEP approximation for a specific type of HQAM, called triangular QAM (TQAM), in AWGN, [6], employs a nearest-neighbors (NNs) method that performs well primarily under high SNR conditions. Although [7] reports a more accurate SEP evaluation for TQAM than [6], the results are still not exact. In [8], the SEP approximation for HQAM employs the average number of NNs and incorporates an adjustment factor to enhance the approach from [6]. In [9], a method for approximating the SEP of HQAM is derived however, compared to [8], the relative error and absolute error (AE) are higher, leading to a decrease in accuracy. Furthermore, in [10], an upper bound and a closed-form approximation for the SEP of HQAM are proposed, along with a low-complexity detection scheme. While [10] it reduces complexity compared to [8] and [9], its accuracy is limited for smaller constellations ($M < 256$).

This ongoing gap between analytical accuracy and computational simplicity motivates the need for a more accurate and computationally efficient SEP approximation, along with a low-complexity detection scheme for HQAM, particularly for smaller constellation orders. This letter offers the following key contributions:

- We develop a new SEP approximation for all HQAM constellations, offering higher accuracy with lower complexity.

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TABLE I
AE FOR 256-HQAM

SNR (dB)	AE Approx. (6)	AE [8]	AE [9]	AE [10]
-4	0.0039	0.0041	0.1306	0.0046
-1	0.0045	0.0056	0.1305	0.0050
2	0.0048	0.0072	0.1287	0.0052
5	0.0051	0.0095	0.1237	0.0058
8	0.0062	0.0132	0.1127	0.0077
11	0.0049	0.0152	0.0872	0.0081
14	0.0053	0.0192	0.0460	0.0110
17	0.0059	0.0221	0.0085	0.0135
20	0.0088	0.0236	0.0458	0.0150
23	0.0086	0.0169	0.0357	0.0093

- A new generalized cluster-based detection scheme is proposed for all HQAM constellations.
- Additionally, we investigate the average SEP (ASEP) performance of HQAM in the presence of imperfect channel state information (CSI) over Nakagami-*m* fading channels.

II. System Model

An *M*-ary HQAM constellation, depicted in Fig. 1(a), is characterized by a set $A_M = \{s_i \in R^2, i = 0, \ldots, M-1\}$ containing M symbols, where A_M is a subset of the infinite grid $\mathcal{G} = \{v \in R^2 : v = n_1 v_1 + n_2 v_2 + z_0\}$ with $v_1 = [d/2, \sqrt{3}d/2]^T$ and $v_2 = [d/2, -\sqrt{3}d/2]^T$ as the basis vectors of the 2D grid. Here, $d > 0$ denotes the smallest distance between points, n_1 and n_2 are integer coefficients, and z_0 is an offset in the 2D plane. The maximum likelihood (ML) decision regions, illustrated in Fig. 1(a) feature central areas that form equilateral hexagons with internal angles of $2\pi/3$. The outer regions may include one or more angles of $2\pi/3$.

Moreover, we can describe the received signal at the destination (receiver) node as

$$r = s + n_0. \tag{1}$$

Here, s represents the transmitted symbol and n_0 denotes AWGN at the receiver, with zero mean and variance $N_0/2$, where $N_0/2$ is the two-sided noise power spectral density [10].

III. SEP Approximation for AWGN Channels

An approximate expression for the SEP of an *M*-ary signal set in an AWGN channel is given by

$$P_e = AQ\left(\frac{d}{\sqrt{2N_0}}\right), \tag{2}$$

where $Q(\cdot)$ represents the Gaussian Q-function, defined as $Q(z) = \frac{1}{\sqrt{2\pi}} \int_z^\infty e^{-\frac{t^2}{2}}\, dt$. Here, $d = \sqrt{2N_0\kappa\gamma_0}$ denotes the NNs distance, and $\gamma_0 = \frac{P_s}{N_0}$ represents the transmit SNR, where P_s is the transmitted power. The parameter $\kappa = \frac{24}{7M-4}$ is defined for various constellation orders, as specified in [1]. Moreover, *A* represents the average number of NNs, calculated as $A = \frac{1}{M}\sum_{i=0}^{M-1} A(i)$, where *A*(*i*) is the count of NNs for the symbol s_i. As an illustration, the simplest form of HQAM, specifically the 3-PSK is depicted in Fig. 1(b). The transmitted symbol s_0 is surrounded by two NNs, s_1 and s_2. Given that s_1 is also a NN of s_2 (and vice versa), s_1 and s_2 are considered a pair of adjacent NN to s_0. The error region in the NN approximation is covered by two overlapping half-planes, leading to an overestimation of the SEP. Therefore, the precise expression for the SEP in 3-PSK is

$$P_{3-PSK} = 2Q(\sqrt{\kappa\gamma_0}) - C, \tag{3}$$

where *C* represents the correction factor, addressing the issue of double counting due to the overlap. Given that 3-PSK is a simplex signal [11], *C* can be expressed as

$$C = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[2Q\left(\sqrt{\kappa\gamma_0}\right) - \{1 - (1 - Q(z))^2\}\right] \times e^{-\left(z-\sqrt{2\kappa\gamma_0}\right)^2/2}\, dz. \tag{4}$$

The calculation of C is provided in Appendix, where

$$C \approx 1.3318 Q\left(\sqrt{\frac{10}{11}\kappa\gamma_0}\right) Q\left(\sqrt{\frac{1}{3}\kappa\gamma_0}\right). \tag{5}$$

In HQAM, several overlaps occur at angles of $2\pi/3$, each associated with a pair of NNs. Furthermore, by applying the correction method, we extend (3) to any HQAM constellation, resulting in a novel and accurate closed-form expression for the SEP of HQAM, given as

$$P_{HQAM} = AQ(\sqrt{\kappa\gamma_0}) - BQ\left(\sqrt{\frac{10}{11}\kappa\gamma_0}\right) Q\left(\sqrt{\frac{1}{3}\kappa\gamma_0}\right), \tag{6}$$

where $B = 1.3318A_c$. The corresponding values of $A = 2(3 - 4M^{-1/2} + M^{-1})$ and $A_c = 6(1 - M^{-1/2})^2$ are given in [1]. The approximation presented in (6) is a novel closed-form expression that has not been previously reported in the literature. It is worth noting that these approximations offer lower complexity for a given SNR compared to existing methods. Additionally, Table I presents the AE of the SEP for the 256-HQAM constellation, comparing the proposed approximation with those reported in [8], [9], and [10]. The AE is defined as $AE = |\,\text{approximated SEP} - \text{exact SEP}\,|$. The proposed approximation in (6) exhibits a lower AE compared to existing methods and offers significantly higher accuracy with reduced computational complexity.

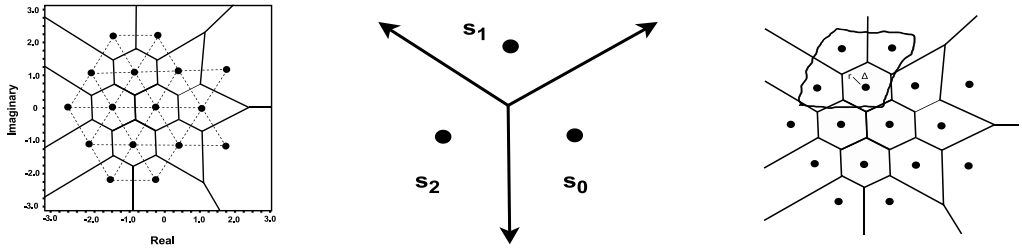
IV. Cluster-Based Detection Scheme

In this section, we propose a new low-complexity detection scheme that achieves performance close to that of maximum likelihood detection (MLD). Fig. 1(c) depicts the proposed detection scheme of a received symbol *r*, and the procedure is summarized in Algorithm 1. Assume the *i*-th symbol of the constellation points is of the form $s_i = x_i + jy_i$, to convert the constellation points into real-valued 2D coordinates, we can express them as

$$D_r = \begin{bmatrix} \mathrm{Re}(s_1) & \mathrm{Im}(s_1) \\ \mathrm{Re}(s_2) & \mathrm{Im}(s_2) \\ \vdots & \vdots \\ \mathrm{Re}(s_M) & \mathrm{Im}(s_M) \end{bmatrix}$$

where *j* is the imaginary unit. We partition D_r into multi-stages clusters $E = \{C_p : p = 1, 2, \ldots, P \le \sqrt{M}\}$ and $D_r = \bigcup_{p=1}^{P} C_p$ [12]. The clusters are mutually exclusive and the constellation points in the cluster are distributed uniformly. We then build a K-D Tree [13] for each cluster of *E* with the cluster-specific constellation points. The Euclidean distance is calculated between the detected symbol and the received symbols *r* from the constellation point of the cluster found via

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(a) (b) (c)

Fig. 1. (a) 16 HQAM Constellation, (b) 3-PSK Constellation, (c) Cluster Based Detection Scheme.

Algorithm 1 Algorithm for Symbol Detection

Require: $r = (x_r, y_r)$
Ensure: Detected symbol $\hat{s}$
1: minimum distance $(\Delta_{min}) \leftarrow inf$
2: best symbol $\leftarrow null$
3: **for** $i \leftarrow 1$ to P **do**
4: detected index = K-NN Search($r, K - DTree_E$)
5: globalindex = clusterIndices(E)(detected index)
6: $\Delta = \|\mathbf{r} - \mathbf{globalindex}\|_2$
7: **if** $\Delta < \Delta_{min}$ **then**
8: $\Delta_{min} \leftarrow \Delta$
9: $bestsymbol \leftarrow globalindex$
10: $\hat{s}$ = best symbol
11: **end if**
12: **end for**
13: **return** best symbol

K-D tree search. Thus, the symbol that is detected corresponds to the cluster with the minimum Euclidean distance (Δ) from all the other clusters.

Finally, since set E has a length of $\sqrt{M}$, the K-D tree built on this set has a time complexity of $O(\log\sqrt{M})$. Thus, the average time complexity of the algorithm is $O(\log\sqrt{M})$ for an M-order constellation. This approach guarantees optimal detection with significantly lower complexity than a full search over the entire constellation. Therefore, using a K-D tree accelerates the NNs search, making the detection of high-order HQAM schemes both computationally efficient and scalable.

V. ASEP Analysis Under Nakagami-m Distributed Environment

We consider a single-hop wireless communication network in which the source and destination operate in half-duplex mode with single antenna, while maintaining symbol-level synchronization. The channel coefficient amplitude $\mathcal{H}$ follows a Nakagami-m distribution, $\mathcal{H} \sim \text{Nak}(m, \Omega_{\mathcal{H}})$. The conventional coherent systems rely on channel estimation to obtain CSI at the receiver [14], [15]. Thus, under imperfect CSI, channel estimation errors arise, and the least mean squared estimate (LMSE) method is commonly used for estimation. The actual channel coefficient vector $\mathcal{H}$ and its estimate $\hat{\mathcal{H}}$ can be related as $\mathcal{H} = \hat{\mathcal{H}} + \mathcal{E}_e$, where $\mathcal{E}_e \sim \mathcal{CN}(0, \Omega_e)$ represents the channel estimation error (CEE), modeled as a complex Gaussian random variable with zero mean and variance Ω_e. The CEE variance is given by $\Omega_e = \frac{\Omega_{\mathcal{H}}}{1+\rho\gamma_0\Omega_{\mathcal{H}}}$ and $\hat{\Omega}_{\mathcal{H}} = \Omega_{\mathcal{H}} - \Omega_e = \frac{\rho\gamma_0\Omega_{\mathcal{H}}^2}{1+\rho\gamma_0\Omega_{\mathcal{H}}}$, where ρ $(0 \le \rho \le 1)$ characterizes the quality of channel estimation. During transmission, the source transmits a signal X with power P_s to the destination, where the received signal is given by

$$Y = \sqrt{P_s}\left(\hat{\mathcal{H}} + \mathcal{E}_e\right)X + n_0 \tag{7}$$

The instantaneous SNR, γ, at the destination can be expressed as $\gamma = \frac{\hat{\gamma}}{1+\xi}$ where $\hat{\gamma} = \frac{P_s|\hat{\mathcal{H}}|^2}{N_0}$ and $\xi = \Omega_e\gamma_o$. Since the wireless link follows a Nakagami-m distribution, the cumulative distribution function (CDF) of γ is expressed as $F_\gamma(\gamma) = \frac{1}{\Gamma(m)}\Upsilon(m, \frac{m\gamma}{\bar{\gamma}})$, where $\Gamma(\cdot)$ and $\Upsilon(\cdot,\cdot)$ represent the Gamma function and the lower incomplete Gamma function, respectively. Here, $\bar{\gamma} = E[\gamma]$, where $E[\,\cdot\,]$ denotes the expectation operator. Furthermore, the ASEP for an HQAM scheme can be mathematically expressed using a well-known CDF-based approach as

$$P_s(e) = -\int_0^\infty P'_{\text{HQAM}}(e|\gamma)F_\gamma(\gamma)\,d\gamma, \tag{8}$$

where $P'_s(e|\gamma)$ denotes the first derivative of the conditional SEP, $P_s(e|\gamma)$, as given in (6). Utilizing the first-order derivative of the error function, $\frac{d}{dz}\text{erf}(z) = \frac{\sqrt{2}}{\sqrt{\pi}}e^{-z^2}$ as stated in [16, (7.1.19)], and the expansion of the error function, $\text{erf}(z) = \frac{\sqrt{2}}{\sqrt{\pi}}ze^{-z^2}\,{}_1F_1(1, \frac{3}{2}, z^2)$, as provided in [16, (7.1.21)], the first-order derivative of (6) can be expressed as

$$P'_s(e|\gamma) = -\frac{A}{2}\sqrt{\frac{\kappa}{2\pi}}\gamma^{-1/2}\exp\left(-\frac{\kappa\gamma}{2}\right) + \frac{B}{4}\sqrt{\frac{\kappa}{\pi}}\gamma^{-1/2}\Big[\sqrt{\frac{5}{11}} \times \exp\left(-\frac{5\kappa\gamma}{11}\right) + \frac{1}{\sqrt{6}}\exp\left(-\frac{\kappa\gamma}{6}\right)\Big] - \frac{B\kappa}{2\pi}\sqrt{\frac{5}{66}} \times \exp\left(-\frac{41\kappa\gamma}{66}\right)\Big[{}_1F_1\left(1; \frac{3}{2}; \frac{\kappa\gamma}{6}\right) + {}_1F_1\left(1; \frac{3}{2}; \frac{5\kappa\gamma}{11}\right)\Big]. \tag{9}$$

Substituting (9) along with $F_\gamma(\gamma)$ into (8), and then solving the required integrals with the aid of [17, (6.455.2), (7.621.4),

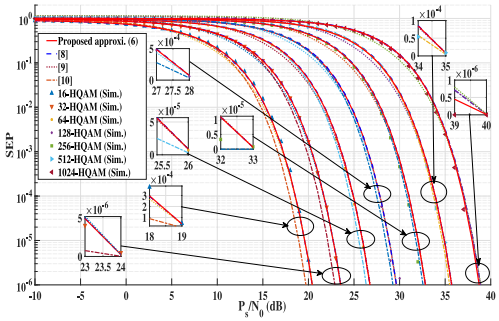
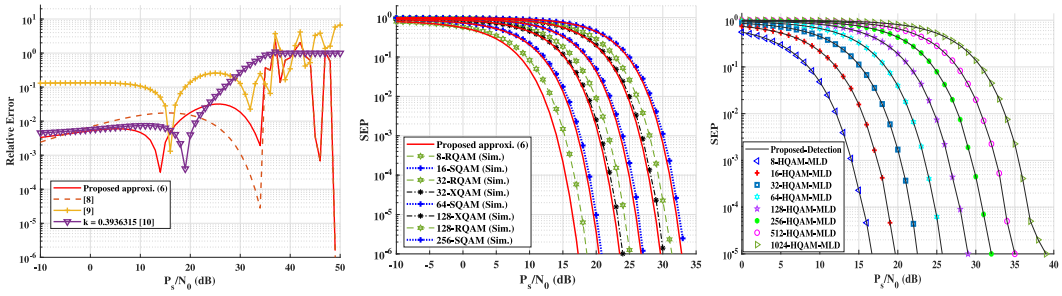


Fig. 2. Compares the SEP approximation versus P_s/N_0.

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(a) (b) (c)

Fig. 3. (a) Relative error versus P_s/N_0 of 256-HQAM, (b) Simulated and theoretical SEP of HQAM Constellations, and (c) SEP versus P_s/N_0 for detected symbol over AWGN.

and (8.352.1)], the ASEP expression for general-order HQAM can be expressed as

$$P_s(e) = \frac{A}{2}\sqrt{\frac{\kappa}{2\pi}}\frac{1}{\Gamma(m)}\hat{F}\left(\frac{m}{\bar{\gamma}_{sd}},0,0,\sqrt{\frac{\kappa}{2}},\sqrt{\frac{\kappa}{2}}\right) - \frac{B}{4}\sqrt{\frac{\kappa}{\pi}} \times \frac{1}{\Gamma(m)}\left[\sqrt{\frac{5}{11}}\hat{F}\left(\frac{m}{\bar{\gamma}_{sd}},0,0,\sqrt{\frac{5\kappa}{11}},\sqrt{\frac{5\kappa}{11}}\right) + \frac{1}{\sqrt{6}} \times \hat{F}\left(\frac{m}{\bar{\gamma}_{sd}},0,0,\sqrt{\frac{\kappa}{6}},\sqrt{\frac{\kappa}{6}}\right)\right] + \frac{B\kappa}{2\pi}\sqrt{\frac{5}{66}}\left[\nabla\left(0,0,\frac{41\kappa}{66},\frac{\kappa}{6}\right) - \sum_{k_1=0}^{m-1}\frac{\left(\frac{m}{\bar{\gamma}_{sd}}\right)^{k_1}}{k_1!}\nabla\left(k_1,\frac{m}{\bar{\gamma}_{sd}},\frac{41\kappa}{66},\frac{\kappa}{6}\right) + \nabla\left(0,0,\frac{41\kappa}{66},\frac{5\kappa}{11}\right) - \sum_{k_2=0}^{m-1}\frac{\left(\frac{m}{\bar{\gamma}_{sd}}\right)^{k_2}}{k_2!}\nabla\left(k_2,\frac{m}{\bar{\gamma}_{sd}},\frac{41\kappa}{66},\frac{5\kappa}{11}\right)\right], \quad (10)$$

where $\hat{F}(\varrho,\sigma,\zeta,\mu,\nu) = \frac{(\varrho)^{m+\sigma}\Gamma(m+1/2+\zeta)}{(m+\sigma)[(\mu^2+\nu^2)/2+(m/\bar{\lambda}_{sd})]^{m+1/2+\zeta}} \times {}_2F_1(1, m+1/2+\zeta; m+1+\sigma; \varrho/[(\mu^2+\nu^2)/2+(m/\bar{\lambda}_{sd})])$, and $\nabla(\mu_0,\nu_0,\eta_0,\rho_0) = \Gamma(\mu_0+1)(\nu_0+\eta_0)^{-(\mu_0+1)} \times {}_2F_1(1,\mu_0+1;3/2;\rho_0/(\nu_0+\eta_0))$.

VI. Simulation Results

In Fig. 2, the approximated SEP derived from (6) for HQAM constellations is shown over an AWGN channel. The figure demonstrates that the approximated and simulated curves closely match, confirming the accuracy of the proposed expression across both low and high SNR values for all HQAM constellation orders. Additionally, a comparison is made between the approximated SEP performance curves obtained from (6) and the previously established approximations reported in [8], [9], and [10]. The simulated results align more closely with the proposed approximation than with the existing ones, further validating accuracy of the proposed method.

Fig. 3(a) illustrates the relative error of the SEP approximation presented in (6) for the 256-HQAM constellation, in comparison with the approximations proposed in [8], [9], and [10]. The relative error quantifies the uncertainty in the approximation relative to the exact SEP value. As evident from Fig. 3(a), the proposed approximation performs well in the SNR range of -6 dB to 18 dB. Beyond this range, the approximation proposed in [8] performs better. At higher SNR values, starting from 34 dB, its performance becomes comparable to that of the approximation in [8]. The proposed method proves highly reliable, particularly in low and high-SNR regions, ensuring superior performance. Furthermore, the proposed approximation demonstrates high accuracy, with the relative error remaining below 10^{-2} (less than 1%) in the SNR range of -10 dB to 18 dB.

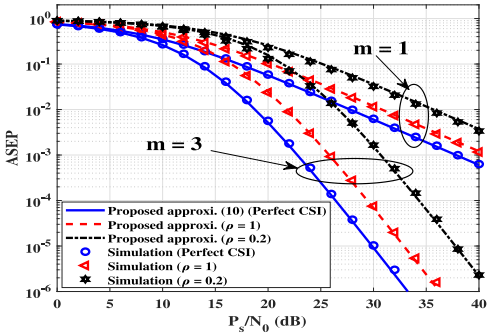


Fig. 4. 16 HQAM with perfect and imperfect CSI.

Fig. 3(b) presents a comparative SEP performance analysis of various QAM schemes over an AWGN channel, highlighting SNR gains at an ASER level of 10^{-5} using approximated results for HQAM and simulated results for the others. The results show that HQAM consistently outperforms RQAM, showing a notable SNR gain of 1.361 dB for 8-RQAM, which further increases to 1.551 and 1.662 dB as the size of the constellation increases to 32 and 128, respectively. Furthermore, the 16-HQAM scheme shows a performance gain of approximately 0.226 dB over 16-SQAM, which increases to around 0.473 dB and 0.786 dB when the constellation order is extended to 64 and 256, respectively. Additionally, HQAM exhibits a significant advantage over XQAM, achieving an approximate gain of 0.507 dB for 32-XQAM and 0.510 dB for 128-XQAM. These improvements can be attributed to the lower peak-to-average power ratio of HQAM than others, resulting from its compact constellation structure and optimized Euclidean distance between signal points.

Fig. 3(c) illustrates the SEP performance of the proposed detection method compared to conventional ML detection across various values of P_s/N_0 and different M. For all HQAM constellation sizes and P_s/N_0 values, the SEP performance of the proposed method matches that of conventional ML detection. This demonstrates that the proposed method achieves a significant reduction in implementation complexity, from $O(M)$ to $O(\log\sqrt{M})$, while maintaining equivalent performance.

Fig. 4 presents the ASEP performance of the 16-HQAM scheme over a Nakagami-m fading channel for both perfect ($\rho \to \infty$) and imperfect CSI ($\rho = 0.2, 1$). The simulation

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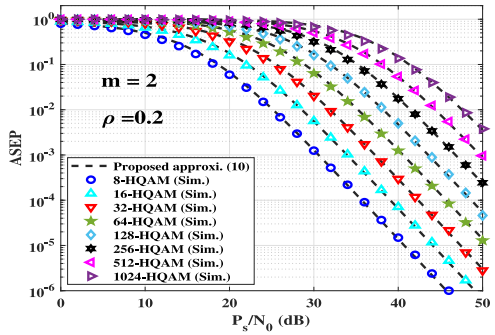


Fig. 5. ASER performance of M-HQAM scheme.

results exhibit strong agreement with the theoretical analysis, validating the derived expression in (10). Additionally, increasing m from 1 to 3 significantly enhances ASEP performance due to reduced fading severity. However, under imperfect CSI conditions, a noticeable degradation in ASEP is observed due to channel estimation errors. For example, at an ASEP of 10^{-2}, approximately 10 dB SNR loss is observed when m decreases from 3 to 1 under perfect CSI. Similarly, for a fixed $m = 3$, an SNR loss of approximately 3 dB and 9 dB is observed for $\rho = 1$ and $\rho = 0.2$, respectively, compared to the perfect CSI case.

Fig. 5 presents a comparison between analytical and simulated results for the ASEP of various HQAM constellations over Nakagami-m fading channels under the assumption of imperfect CSI. As the constellation size increases, a degradation in ASEP performance is observed. This is due to the reduced spacing between constellation points, making the system more vulnerable to noise and channel impairments.

VII. Conclusion

In this letter, we proposed a new, accurate, closed-form approximation for the SEP of general-order HQAM constellations and validated its accuracy by comparing it with simulation results over both AWGN and Nakagami-m fading channels. The proposed SEP analysis is more straightforward and accurate compared to the methods in [8], [9], and [10] and extendable to other fading scenarios. Additionally, the proposed detection scheme achieves an average time complexity of $O(\log\sqrt{M})$ while delivering performance identical to that of conventional ML detection.

Appendix
Calculation of C

From (4), we can represent expression for C as

$$C = J_1 - J_2 + J_3, \tag{11}$$

where J_1, J_2, and J_3 can be obtained below as:

$$J_1 = \frac{2}{\sqrt{2\pi}} Q(\sqrt{\kappa\gamma_0}) \int_{-\infty}^{\infty} e^{-\left(z-\sqrt{2\kappa\gamma_0}\right)^2/2} dz, \tag{12}$$

Utilizing [17, (3.323.2)], (12) can be simplified as

$$J_1 = 2Q(\sqrt{\kappa\gamma_0}). \tag{13}$$

$$J_2 = \frac{2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} Q(z) e^{-\left(z-\sqrt{2\kappa\gamma_0}\right)^2/2} dz, \tag{14}$$

With the aid of [18, (B.7)], (14) can be simplified as

$$J_2 = 2Q(\sqrt{\kappa\gamma_0}). \tag{15}$$

$$J_3 = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} Q^2(z) e^{-\left(z-\sqrt{2\kappa\gamma_0}\right)^2/2} dz, \tag{16}$$

By making use of $Q(z) \approx 0.49 e^{-\frac{8}{13}z}\, e^{-\frac{z^2}{2}}$ [19, (17)], (16) can be approximated as

$$J_3 \approx \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} 0.49 e^{-\frac{8}{13}z}\, e^{-\frac{z^2}{2}}\, Q(z)\, e^{-\left(z-\sqrt{2\kappa\gamma_0}\right)^2/2} dz, \tag{17}$$

By using [18, (B.7)], (17) can be written as

$$J_3 \approx \frac{1.8835}{\sqrt{2}} 0.49 e^{\left(-\frac{8}{13}\sqrt{\frac{10}{11}\kappa\gamma_0} - \frac{5}{11}\kappa\gamma_0\right)} Q\left(\sqrt{\frac{1}{3}\kappa\gamma_0}\right) \tag{18}$$

Finally, with the aid of [19, (17)], (18) can be reformulated in the form of $Q(\cdot)$ function as

$$J_3 \approx 1.3318 Q\left(\sqrt{\frac{10}{11}\kappa\gamma_0}\right) Q\left(\sqrt{\frac{1}{3}\kappa\gamma_0}\right). \tag{19}$$

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